

### (a) Twin construction

A: dog    B: rooster



Reverse query  $\leftrightarrow$  swap target

$q_+$ : dog followed by rooster  $\rightarrow W_+$

$q_-$ : rooster followed by dog  $\rightarrow W_-$

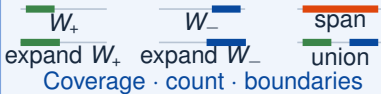
### (b) RelTwin training

#### Stage 1: Answer learning

|       | $y(W_+)$              | $y(W_-)$              |
|-------|-----------------------|-----------------------|
| $q_+$ | $s_{++}$ $\uparrow$   | $s_{+-}$ $\downarrow$ |
| $q_-$ | $s_{-+}$ $\downarrow$ | $s_{--}$ $\uparrow$   |

Competition + reversal consistency

#### Stage 2: Set preferences



**Inference** Audio + query  $\rightarrow$  Adapted backbone  $\rightarrow$  Timestamps

### (c) Window selection

